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Vision and self-reported mobility performance in patients with low vision

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Background: As vision plays a significant role in mobility performance, it is usual to refer low vision patients, particularly those who complain of mobility difficulties, for orientation and mobility (O&M) training to help them maintain safe independent travel. Our study aimed to determine whether there was a relationship between measures of vision and self-reported mobility, and the applicability of a patient-based mobility assessment in patients with heterogeneous causes of visual loss.

Method: We assessed the high and low contrast visual acuity, visual field and scanning ability of 30 patients with low vision. A validated mobility questionnaire was used to assess the participants' perceived visual ability for independent mobility.

Results: Vision was significantly correlated with self-reported mobility performance, however, visual field was a significant predictor accounting for 56.5 per cent of the variance. The instrument was well constructed with valid content and high reliability scores.

Conclusions: Self-reported mobility performance together with measures of vision could be used as a guide to refer patients for O&M training. The patient-based assessment instrument was valid to measure perceived visual ability for independent mobility in patients with heterogeneous causes of visual loss.

Key words: low vision, orientation and mobility training, scanning ability, self-reported mobility, visual acuity, visual field

Vision plays an important role in navigation of the environment. Research has shown that visual acuity, visual field and contrast sensitivity correlate with mobility performance, however, visual field and contrast sensitivity are stronger predictors than visual acuity.[1–4]

In a group of participants with heterogeneous causes of low vision, Marron and Bailey[1] showed that visual field and contrast sensitivity together accounted for 53 per cent of the variance in mobility performance. In a more recent study, Kuyk, Elliott and Fuhr[3] investigated mobility performance (time to complete a course and the number of mobility incidents) of a large sample of visually impaired adults (n = 156) using measures of visual sensory and perceptual functions. They found that visual field, scanning reaction time and contrast sensitivity together accounted for 45 per cent and 48 per cent of the variance in total time and total errors, respectively.

Using patients with age-related macular degeneration (AMD), Brown and associates[5] demonstrated that visual acuity was a significant predictor accounting for more than 70 per cent of the variance in each mobility measure.

Thus, if there are deficits in these visual functions, one's navigational skills would be adversely affected and consequently, one may not be able to travel independently.

Fortunately, low vision patients can be referred by their eye care practitioners and/or rehabilitation officers for orientation and mobility (O&M) training to maintain their ability to travel indepen-

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dently.[6] In O&M training, low vision patients learn new mobility skills to ambulate and negotiate the environment safely and independently.[7] By learning new skills to compensate for their visual loss, low vision patients have increased travel options, thus maintaining their quality of life. Geruschat and Del'Aune[8] showed that mobility skills of low vision patients improved significantly following O&M training. Further, Kuyk and associates[9] showed that self-reported mobility function of low vision patients improved significantly following a comprehensive blind rehabilitation program that included O&M training.

There are no established guidelines in New Zealand or Australia that eye care practitioners and rehabilitation officers could use to refer patients for O&M training. In Australia, low vision patients are usually referred for general vision rehabilitation services, which may include O&M training based on their visual field and visual acuity.[10,11] Self-reported mobility difficulty is often used as a general guide by eye care practitioners and rehabilitation officers to determine if O&M training is warranted. This was confirmed by Beggs[12] who found that mobility officers at a residential rehabilitation centre referred their clients for O&M training based on how the clients felt about their mobility skills. It is not surprising that low vision people's perception of their own mobility skills will determine whether they are confident to travel independently. It may not be unexpected that those patients with poor vision are likely to report difficulties with travelling independently.

Our study sought to determine if there is a relationship between vision and self-reported mobility performance. We hypothesised that vision is significantly related to self-reported mobility performance.

We used a validated patient-based assessment instrument, which was used in patients with retinitis pigmentosa[13] or glaucoma[14] to assess self-reported mobility performance. Our study also aimed to determine the applicability of the questionnaire in patients with heterogeneous causes of low vision.

ID	Age (years)	Ocular condition	Visual acuity
1	85	AMD	6/120
2	53	Retinitis pigmentosa	6/9.5
3	41	Choroideraemia	6/19
4	31	Leber's optic atrophy, glaucoma	6/6000
5	81	AMD	6/95
6	82	AMD	6/30
7	82	AMD	6/48
8	49	Ocular trauma	6/6000
9	40	Congenital glaucoma	6/24
10	85	Hypentensive retinopathy	6/38
11	27	Glaucoma, Peter's anomaly	6/120
12	73	AMD	6/950
13	48	Retinitis pigmentosa	6/75
14	26	Retinitis pigmentosa	6/75
15	85	AMD	6/12
16	77	AMD	6/12
17	64	Central retinal vein occlusion	6/15
18	83	AMD	6/38
19	86	AMD	6/9.5
20	75	AMD	6/15
21	79	AMD	6/12
22	81	AMD	6/12
23	81	AMD	6/12
24	91	Diagnosis unknown	6/7.5
25	75	Cataracts	6/24
26	83	Glaucoma	6/15
27	83	Macular hole	6/9.5
28	76	Diagnosis unknown	6/75
29	44	Retinitis pigmentosa	6/15
30	48	Choroidal scar	6/5

AMD: age-related macular degeneration

Table 1. Characteristics of subjects

METHODS

Participants

Thirty-two low vision participants were recruited from the Low Vision Clinic, the University of Auckland. Two participants were excluded from the study because they were no longer visually impaired due to recent cataract surgery. The participants' ages ranged from 26 to 91 years and they had various ocular conditions (Table 1). The mean age (standard deviation) of the participants was 65.1 (± 21.7) years.

The Auckland Ethics Committee of the New Zealand Ministry of Health granted approval for the study. All participants gave written informed consent for participation following explanation of the nature of the study.

Measurements of vision

The participants' vision was assessed binocularly with their habitual optical corrections using the following visual function tests.

HIGH CONTRAST VISUAL ACUITY (HCVA)

Binocular HCVA was measured using the high contrast Bailey-Lovie distant chart, scored letter-by-letter and expressed as the logarithm of the minimum angle of resolution.[15]

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LOW CONTRAST VISUAL ACUITY (LCVA)

LCVA was measured using the low contrast Bailey-Lovie distant chart and scored using the same letter-by-letter method as the high contrast Bailey-Lovie chart.[15]

There were 10 participants (33 per cent) who were unable to perceive any letters from the low contrast chart irrespective of the test distance. This indicates that using the low contrast Bailey-Lovie chart for contrast sensitivity assessment may not be appropriate for low vision patients who have severe visual loss.

VISUAL FIELD (BVF)

The kinetic program of the Allergan Humphrey Field Analyser (600 series) was used to measure the participants' visual field with the following stimulus parameters used by Soong, Lovie-Kitchin and Brown:[16]

1. size: IV 4 Ē (Goldmann notation)
2. diameter: 4.6 mm
3. luminance: 13.2 cd/m^2
4. background: 10 cd/m^2
5. speed: 4 deg/sec.

A C++ Builder computer program was used to convert polar co-ordinates of the isopters into points on the surface of a sphere to compute the participants' visual field in percentage of a sphere.[17] The remaining binocular visual field as a percentage of a sphere (BVF) was calculated by taking the difference between the area of the total visual field and that of any scotoma within the field area.

SCANNING ABILITY (SA)

The participants' SA were assessed by measuring their scanning reaction time (SRT),[18] using the Director 8 Macromedia computer program based on the time (seconds) taken to locate, in order, a sequence of four numbers (1, 2, 3, 4) on each of four black and white photographs with randomly scattered distractor stimuli (Figure 1).

In each photograph, the numbers were located at different positions so that the participants were required to scan the computer monitor to locate them. The presentations of the four photographs were randomised among participants to avoid any bias in performance on a photograph.



Figure 1. Black and white photographs, each with four numbered targets for measurement of the participants' scanning reaction time (SRT). Numbers are not drawn to scale.

Each photograph was displayed on a 17-inch computer monitor (average luminance, 56.2 cd/m^2) with the subject seated to provide a 40 cm viewing distance to the monitor. The participants used their appropriate spectacle correction for the distance. The size of each number was equivalent to 6/120.

Each participant was required to indicate the location of the numbers on each photograph by using a plastic pointer. The response was keyed into the program by an experimenter using the computer keyboard. There were five trials for each photograph, thus totalling 20 trials for each subject for the four photographs. There were no significant differences in performance between the photographs and between trials, when the scanning ability test was used with a group of normally sighted participants (L Wood, M Collie, M Hay and GP Soong, personal communication). We calculated the mean of the five trials for each photograph, then the mean of the four photographs to determine the SRT of each participant.

Two participants were unable to perceive any of the numbered targets on the photographs due to severe visual loss.

The participants' scanning ability, SA (per second) was calculated as: SA = 1/SRT.

We assigned a nominal value of five standard deviations about the mean to those participants who were unable to perceive letters and numbers on the Bailey-Lovie charts and on the photographs in the scanning ability assessment, respectively. For the participants who were unable to perceive the stimulus of the kinetic perimetric program, we assigned a value of zero to each participant's BVF score, indicating that there was no remaining visual field recorded.

Assessment of self-reported mobility

We used the patient-based assessment instrument, the Independent Mobility questionnaire to assess the participants' perceived visual ability for independent mobility.[13,14] The questionnaire was administered verbally by an experimenter. Using Rasch analysis, Turano and as-

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sociates[13] and Turano, Massof and Quigley[14] showed that the instrument was well constructed with valid content and high reliability for assessing perceived visual ability for independent mobility in patients with retinitis pigmentosa[13] and patients with glaucoma.[14] There is no other validated patient-based mobility assessment instrument that could be used to assess self-reported mobility of patients with heterogeneous causes of visual loss.

The instrument consisted of 35 items of mobility situations (part one) and a series of questions requiring 'Yes' or 'No' answers about the participants' mobility-related behaviour, such as the history of falls and whether they had had mobility training (part two). The participants were required to rate on a scale of 1 to 5 the level of difficulty they encountered in each of the 35 mobility situations, when unaccompanied by a person or without the use of a mobility device. They were informed that 1 meant 'no difficulty' and 5 meant 'extreme difficulty' while 2, 3 and 4 were not explained. The participants were also told to score a mobility situation 'NA' if they performed an activity with assistance and to score 'X' if the difficulty was due to causes other than visual loss. An abbreviated form of the questionnaire is included in Table 3. The complete questionnaire can be viewed in the appendix of Turano and associates.[13]

Data analysis

The Statistical Package for the Social Sciences (SPSS) was used for data analysis. Using the Kolmogorov-Smirnov test, we found that the measures of vision were not significantly different from a normal distribution ($p > 0.05$).

We also used Rasch analysis to estimate interval measures of perceived visual ability for independent mobility (person measure) and visual ability required for each mobility situation (item measure) from the ordinal ratings of difficulty of the 35 mobility situations.[13,19] The analysis transformed the raw ordinal data into an equal–interval scale using logarithmic transformations of the raw data and probabilistic equations.[20] The measurement unit of the interval scale was the logit, which was used for the person and item measures.

If a person's perceived visual ability for independent mobility was more than the visual ability required for a mobility situation, we expected that he/she had a greater likelihood of reporting less difficulty with the mobility situation. On the other hand, if a person had less perceived visual ability, we expected that they would report greater difficulty in that mobility situation.

Using a Winsteps Rasch-model computer program,[21] an unconditional maximum likelihood estimation routine was performed to determine the parameters of the Wright and Masters[22] version of the Rasch model for rating scale data. The analysis provided reliability indices to indicate the spread of person ability and item difficulty along the measurement continuum.[20] The person reliability index indicated the replicability of person ordering using a different instrument measuring the same construct, perceived visual ability for independent mobility. The item reliability index indicated the replicability of item placements along the item difficulty continuum if the instrument was applied to a different sample of participants with comparable ability.

In addition, the Rasch measurement model produced chi-square fitting statistics to determine how well the data set met the requirements of the model. The infit value provided information about an item performance for participants whose ability was close to the item's difficulty. The infit statistic was calculated by weighting each squared standardised residual value by its variance for each item responded by every participant and then summed. The total value was then divided by the sum of the variances. The outfit value, which was not weighted, provided information on the influence of outlying scores. The outfit statistic was obtained based on the sum of squared standardised response residuals, that is, for every participant, each standardised residual cell was squared for every item responded to by the participant and summed. The average was then derived by dividing the number of items responded to by the participant. Both outfit and infit statistics were reported in mean squares and were transformed into the standardised forms.

As the person measure estimate was not significantly different from a normal distribution ($p > 0.05$), we used Pearson product moment correlation to determine the relationships between the visual functions and person measure estimate.

	HCVA (LogMAR)	LCVA (LogMAR)	BVF (percentage of a sphere)	SA (per second)
Person measure				
Pearson correlation	0.410	0.576	-0.762	-0.662
p value	0.024	0.001	0.000	0.000

HCVA: high contrast visual acuity
LCVA: low contrast visual acuity (sample size was reduced in the analysis)
BVF: binocular visual field
SA: scanning ability

Table 2. Pearson correlations between person measure (perceived visual ability for independent mobility) and HCVA, LCVA, BVF and SA

RESULTS

Relationship between vision and self-reported mobility performance

The participants' person measure was significantly correlated with HCVA, LCVA, BVF and SA (Table 2). This suggests that patients with reduced HCVA, LCVA, BVF

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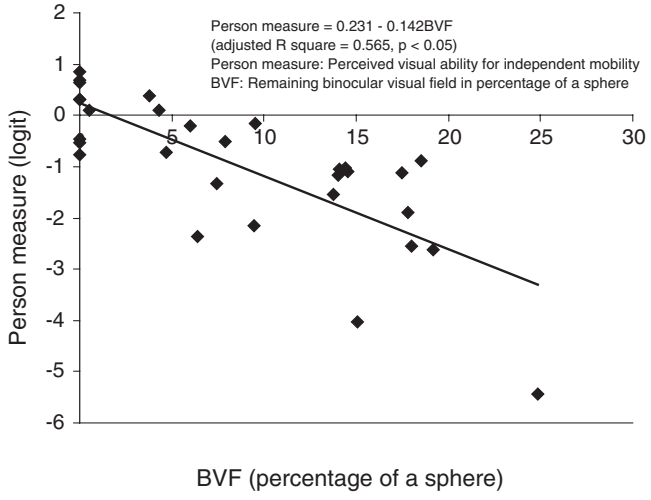


Figure 2. Relationship between person measure and visual field (BVF) with linear regression line

and/or SA are likely to report difficulty with independent mobility.

We also performed a linear stepwise multiple regression to determine the amount of variability in the person measure that can be accounted for by the visual function measures. The analysis showed that BVF was the only significant predictor accounting for 56.5 per cent of the variance in person measure (Figure 2).

Rasch analysis

The Rasch analysis produced a map of the person and item measures, which displayed the level of perceived visual ability for independent mobility (person measure) (left side) and visual ability required for each mobility situation (item measure) (right side) (Figure 3). The items of mobility situations were indicated by the item numbers from the questionnaire, while each subject's perceived visual ability for independent mobility was represented by an 'x'. The logit scale, which had equal values along the vertical axis, was displayed in the centre of the map.

The item-person map showed the relationship between the item measure estimate and person measure estimate. The participants' perceived visual ability locations were plotted, so that any subject who was located at the same position as a mobility situation on the logit scale had a 50 per cent probability of succeeding in performing that mobility situation. If a mobility situation were located further up the scale, the subject would have a higher probability of succeeding in performing the mobility situation.

The participants with least perceived visual ability for independent mobility were at the top of the scale, while those with most perceived visual ability were at the bottom of the scale. The mobility situations that were located further down the scale became more difficult to perform. For example, moving about at work (item 4) was more difficult to perform than moving about at home (item 3) (Figure 3).

Table 3 shows the 35 mobility situations listed in order of least to most required visual ability, together with the error and fit statistics. The mobility items that required the least visual ability were 'moving about in the home', 'moving about at work' and 'walking in familiar areas' while 'walking in high glare areas', 'walking at night' and 'walking in unfamiliar areas' required the most visual ability. Walking in high-glare areas (item 18) was the most difficult to perform requiring the most amount of visual ability.

The mean of the item measure was located at zero by default.[20] The standard deviation for the item measure was close to 1 (standard deviation = 0.91), indicating that the majority of the items were located between +1 and -1 logits (Figure 3). The reliability of the item measure estimates was high on a scale from 0 to 1 (reliability = 0.90), indicating that a large amount of the variance of the item measure was due to the variance in the measure and minimal due to measurement error.

Figure 4 shows the distribution of the person measure estimates. The mean of the person measure estimate of -0.85 was moderately close to zero, indicating that the sample's perceived visual ability was closely matched to the difficulty of the mobility items. In comparison to the item measure, the person measure estimates had a slightly greater spread of variation (standard deviation = 1.13). The reliability was high (reliability = 0.93), indicating that the same ordering of person measure estimates was likely to be expected, if another set of mobility items that measured the same construct (perceived visual ability for independent mobility) were applied to our sample of participants.

In our sample, the means of the unstandardised fitting statistics for both the person measure and item measure estimates were very close to the expected Rasch-model value of 1, hence, producing standardised fitting statistics (Z values) of the expected mean zero in both measures (Table 4). The standard deviations were close to the expected 1.

With the exception of one participant (ID 30), whose values exceeded the Rasch model's expectation by more than two standard deviations, the person and item performance in our study were close to the expected Rasch model for measuring perceived visual ability for independent mobility.

DISCUSSION

In agreement with our hypothesis, we showed that measures of vision were sig-

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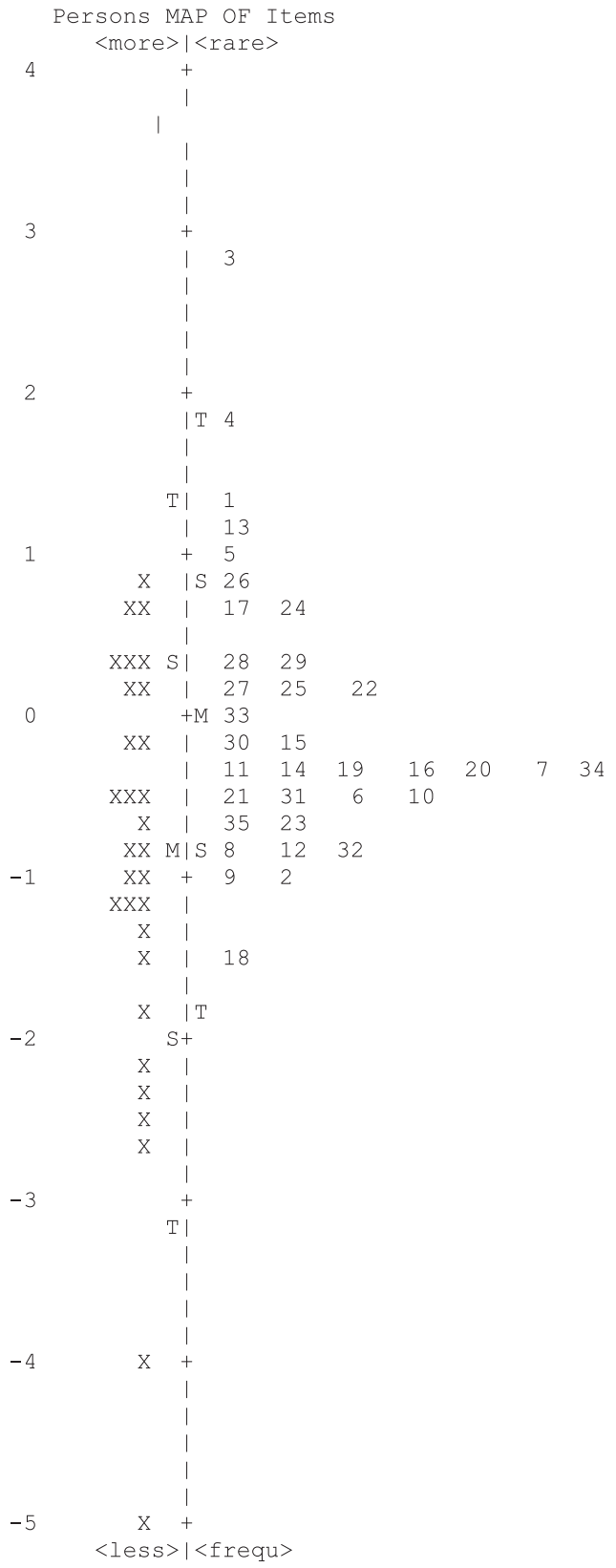


Figure 3. Map of person and item measures of the 30 participants and 35 items of mobility situations

nificantly correlated with perceived visual ability for independent mobility. The current findings confirm previous research, which showed that visual acuity, visual field, contrast sensitivity and scanning ability were important in mobility performance of low vision patients.[2,4,23–25] This indicates that patients with reduced visual acuity, visual field, contrast sensitivity and/or scanning ability are likely to report difficulty with independent mobility, hence may need O&M assessment.

Our findings agree with West and colleagues,[26] who found significant associations between self-reported mobility limitations and high and low contrast visual acuity, contrast sensitivity and visual field in a large population-based study. Szlyk and co-workers[27] also showed that self-reported mobility was significantly correlated with visual field and high contrast visual acuity in a group of patients with retinitis pigmentosa (RP).

Similarly, in a group of participants with RP, Turano and associates[13] showed that perceived visual ability for independent mobility was covariant with contrast sensitivity and visual field but not with high contrast visual acuity. In another study, Turano, Massof and Quigley[14] found that visual acuity was significantly correlated with perceived visual ability in a group of participants with glaucoma.

The difference in findings may be attributed to the different causes of low vision in the studies and the stage or severity of the disease. This suggests that the importance of different visual functions in mobility performance may be dependent on the type and progression of the ocular disease.

We found that visual field was the significant predictor in perceived visual ability for independent mobility in a multivariate regression model, that is, visual acuity, contrast sensitivity and scanning ability did not explain any more of the variance than visual field. Similarly, West and coworkers[26] showed that visual field was a significant predictor of self-reported mobility limitations in their cohort of non-institutionalised residents. By assessing mobility performance, Lovie-Kitchin and associates[17] and Haymes and associates[28]

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	Mobility situation	Item measure (logit)	Error	Infit		Outfit	
				MNSQ	Z_{STD}	*MNSQ*	Z_{STD}
3	Moving in home	3.23	0.95	0.73	0.2	0.21	0.20
4	Moving at work	1.76	0.45	0.90	0.1	0.60	-0.20
1	Walking in familiar areas	1.29	0.29	0.61	-0.9	0.86	0.10
13	Walking up steps	1.18	0.28	1.09	0.4	2.27	1.40
5	Moving in classroom	1.04	0.37	0.70	-0.5	0.66	-0.40
26	Avoid bumping into walls	0.88	0.24	0.54	-1.5	0.49	-0.70
17	Walking through doorways	0.72	0.23	0.50	-1.8	0.63	-0.50
24	Aware of person's presence	0.72	0.23	0.89	-0.2	0.89	0.00
28	Avoid bumping into shoulder-height objects	0.39	0.21	0.78	-0.7	0.81	-0.20
29	Avoid bumping into waist-height objects	0.31	0.20	0.80	-0.6	0.70	-0.50
25	Avoid bumping into people	0.15	0.20	0.67	-1.3	0.53	-1.00
22	Adjusting to lighting changes at night: street lights to indoor	0.14	0.23	1.00	0.1	0.82	-0.10
27	Avoiding bumping into head-height objects	0.11	0.20	1.13	0.6	0.97	0.10
33	Moving around in social gatherings	-0.08	0.20	1.18	0.7	1.09	0.30
15	Stepping onto curbs	-0.10	0.20	0.65	-1.3	0.81	-0.30
30	Avoiding knee-height objects	-0.12	0.19	1.47	1.6	1.13	0.40
14	Walking down steps	-0.26	0.20	1.07	0.3	1.13	0.40
7	Moving about outdoors	-0.27	0.20	1.12	0.5	1.01	0.20
19	Adjusting to lighting changes during the day: indoor to outdoors	-0.30	0.20	1.21	0.8	1.43	1.00
11	Detecting ascending stairs	-0.33	0.19	1.25	1.0	1.03	0.20
16	Stepping off curbs	-0.33	0.19	0.86	-0.4	0.94	0.00
20	Adjusting to lighting changes during the day: outdoors to indoors	-0.38	0.19	1.08	0.4	0.87	-0.20
34	Finding restrooms in public places	-0.42	0.20	1.39	1.3	1.16	0.50
10	Using public transportations	-0.44	0.21	1.21	0.8	1.16	0.60
6	Moving about in stores	-0.48	0.19	1.11	0.5	0.88	-0.20
21	Adjusting to lighting changes at night: indoors to street lights	-0.56	0.21	0.82	-0.6	0.93	0.00
31	Avoiding bumping into low lying objects	-0.57	0.18	1.20	0.8	1.04	0.20
23	Walking in dimly-lit indoor areas	-0.65	0.20	1.38	1.3	2.02	2.10
35	Seeing cars at intersections	-0.67	0.18	1.45	1.6	1.83	2.00
32	Avoiding tripping over uneven travel surface	-0.80	0.19	0.72	-1.1	0.79	-0.50
12	Detecting descending stairs	-0.81	0.18	0.88	-0.4	0.72	-0.70
8	Moving about in crowded situations	-0.84	0.19	1.00	0.1	0.78	-0.50
2	Walking in unfamiliar areas	-0.92	0.20	0.45	-2.4	0.42	-1.90
9	Walking at night	-1.01	0.21	1.25	0.9	0.90	-0.10
18	Walking in high glare areas	-1.58	0.21	0.78	-0.7	0.74	-0.60

MNSQ: mean square fit statistics
Z_{std}: normalised fit statistics
Numbers adjacent to each item indicate the order of the items on the questionnaire

Table 3. Results of Rasch analysis for the 35 questionnaire items applied to the participants

demonstrated that visual field played a more important role than other visual functions, such as visual acuity and contrast sensitivity for predicting mobility performance in people with low vision. This suggests that patients with visual field loss are likely to report mobility problems as a result of not being able to navigate safely in the environment. Therefore, it is essential to refer patients with visual field loss for O&M training when they complain of mobility difficulty, so that they are able to maintain safe independent travel.

Use of instrument in patients with heterogeneous causes of low vision

Our study showed that the patient-based assessment instrument had good content and valid construction in our sample of participants with heterogeneous causes of low vision. This suggests that the recorded observations using the instrument reflected the measurement of the underlying variable, perceived visual ability for independent mobility. Based on our sample of participants, the instrument had high reliability indices indicating both item and person measures were widely spread along the visual ability dimension. Our findings agree with Turano and associates[13] and Turano, Massof and Quigley,[14] who used the instrument on patients

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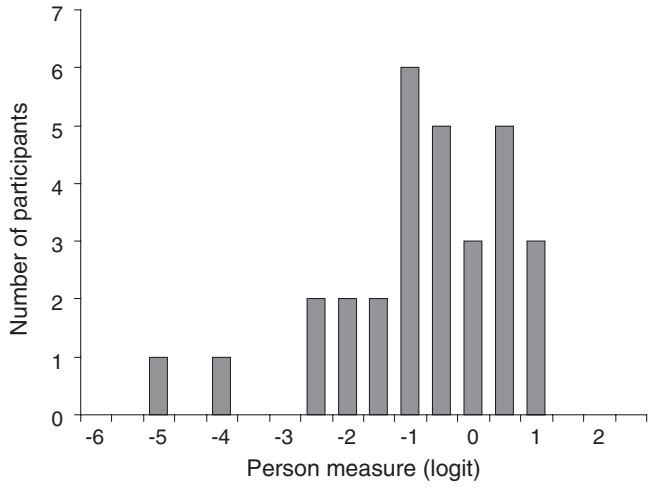


Figure 4. Distribution of person measure estimates derived from Rasch analysis applied to the difficulty ratings of the 35 mobility situations

Person measure (logit)		
	Infit mean square	Outfit mean square
Mean	0.98	0.96
SD	0.39	0.35
	Infit standardised Z	Outfit standardised Z
Mean	0	0
SD	1.3	0.8
Item measure (logit)		
	Infit mean square	Outfit mean square
Mean	0.97	0.95
SD	0.27	0.41
	Infit standardised Z	Outfit standardised Z
Mean	0	0
SD	1.0	0.8

SD: standard deviation

Table 4. Fit statistics of the item measure and person measure estimates

with retinitis pigmentosa or glaucoma to assess perceived visual ability for independent mobility.

Similar to these workers,[13,14] we found that moving about at home and walking in familiar areas required the least visual ability. This is not a surprising result, as low vision people are usually familiar with their home environments and other environments, with which they have had travel experience. There are usually no major changes in object positioning such as furniture at home, thus they are able to anticipate where objects are positioned to avoid bumping into them.

In contrast, the participants in our study had the most difficulty with moving about in dim lighting conditions and extreme lighting conditions such as glare. Turano and associates[13] and Turano, Massof and Quigley[14] also showed that the mobility situations with which their participants had the most difficulty were related to low or extremely high light level conditions. The current findings confirmed the findings of Soong,[29] who showed that mobility performance of low vision people was adversely affected in glare conditions. Similarly, in a forced-choice survey, Smith, Del'Aune and Geruschat[30] showed that walking under glare conditions presented the most difficult mobility task for people with low vision.

Future research

The correlation between contrast sensitivity and perceived visual ability for independent mobility might have been limited by the fact that one-third of the participants were unable to perform the test. Thus, in a future study, contrast sensitivity could be assessed using the Pelli-Robson chart to obtain measurements in participants with severe visual loss. Further, the measurement of scanning ability might have been confounded by a motor component (eye-hand co-ordination), as the participants were required to point at a number using a pointer as soon as they located each number on the computer monitor screen.

We had a small sample of participants with large variability in measures of vision. For future research, a larger sample size could be used to confirm the findings of this study. Further, it would be interesting to determine the correlation between self-reported mobility and objective measures of mobility performance.

It is also essential to determine the critical level of perceived visual ability for independent mobility and measures of vision, at which mobility performance becomes sufficiently impaired to warrant O&M training. This would provide accurate guidelines to eye care practitioners and rehabilitation officers to assess better the need for O&M training for people with low vision. This would enable rehabilitation resources to be used effectively and efficiently in view of the limited financial funding available in Australia and New Zealand for rehabilitation of people with low vision.

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